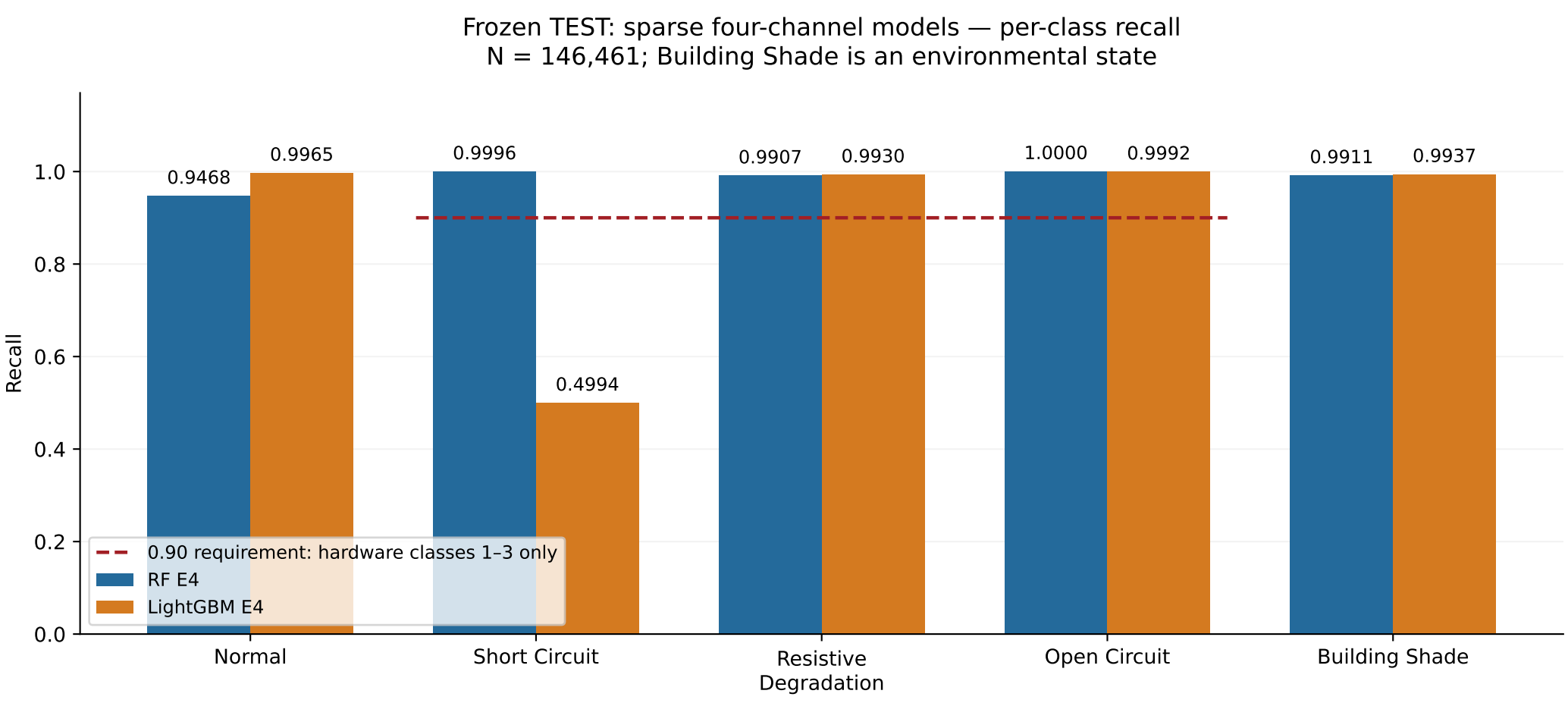
Frozen TEST: sparse four-channel models — per-class recall
N = 146,461; Building Shade is an environmental state
Recall
1.0
0.8
0.6
0.4
0.2
0.0
0.9468
0.9965
0.9996
0.4994
0.9907
0.9930
1.0000
0.9992
0.9911
0.9937
0.90 requirement: hardware classes 1–3 only
RF E4
LightGBM E4
Normal
Short Circuit
Resistive Degradation
Open Circuit
Building Shade